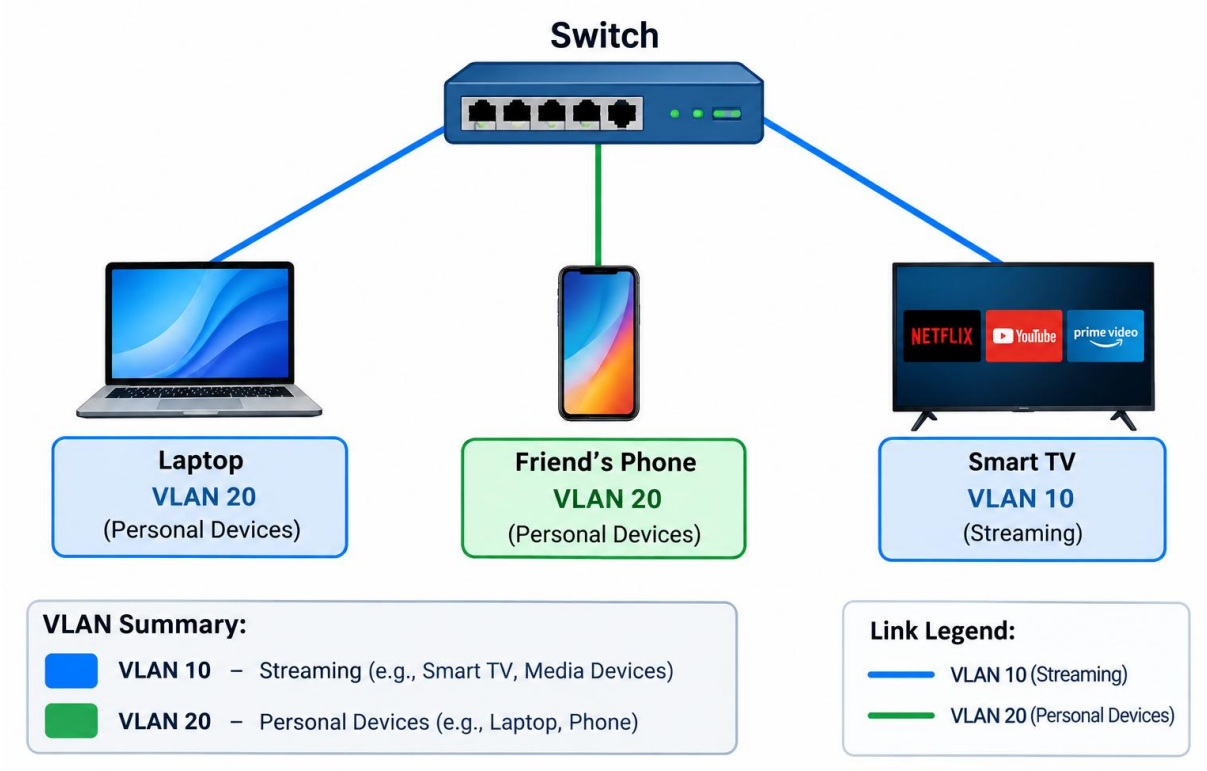
Assignment – 1.10

Switching and VLANS

Q - 1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



Q - 2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

➢ VLAN Configuration:

Aim: Create VLAN 10 and VLAN 20 and check that PCs in different VLANs cannot ping each other.

1. Connect the devices:

- Connect 5 PCs to one Cisco switch.
- PC0, PC1, PC2 → VLAN 10
- PC3, PC4 → VLAN 20

2. Enter these commands in Switch CLI:

Create VLANs:

```
enable
configure terminal
vlan 10
name PERSONAL
exit
vlan 20
```

name STREAMING

exit

Assign 3 PCs to VLAN 10:

interface range fa0/1-3

switchport mode access

switchport access vlan 10

exit

Assign 2 PCs to VLAN 20:

interface range fa0/4-5

switchport mode access

switchport access vlan 20

exit

end

3. Set IP addresses:

PC	IP address
PC0	192.168.10.1
PC1	192.168.10.2
PC2	192.168.10.3
PC3	192.168.20.1
PC4	192.168.20.2

Subnet mask for all PCs: 255.255.255.0

4. Check VLANs:

Enter:

show vlan brief

5. Test ping:

- PC0 to PC1 → **Ping successful**
- PC3 to PC4 → **Ping successful**
- PC0 to PC3 → **Ping failed**

Q - 3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

- **Move a PC from VLAN 10 to VLAN 20:**
- **Step 1: Before configuration:**

  PC1 is connected to switch port Fa0/1 in VLAN 10.

  Switch> enable

  Switch# configure terminal

  Switch(config)# interface fa0/1

  Switch(config-if)# switchport mode access

  Switch(config-if)# switchport access vlan 10

- **Step 2: Move PC1 to VLAN 20:**

  Switch(config)# interface fa0/1

  Switch(config-if)# switchport access vlan 20

  Switch(config-if)# end

- **Step 3: Check the configuration:**

  Switch# show vlan brief

- **Step 4: Test connectivity:**

  - Ping a PC in VLAN 20: should succeed if configured correctly.
  - Ping a PC in VLAN 10: should fail if inter-VLAN routing is not configured.

Q - 4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

- **Assign 10 Switch Ports to VLAN 30:**
- **Simulator:** Cisco Packet Tracer
- **Commands:**

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 30
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# end
Switch# show vlan brief
```

- **Result:** All 10 ports (Fa0/1 to Fa0/10) are assigned to VLAN 30.